

Protocol Analysis Report

Controlled Lab Network Traffic | 7 October 2026

1. Executive Summary

This report analyzes the network protocols observed in a controlled lab consisting of a Windows victim host (192.168.40.1) and a Kali Linux analysis host (192.168.40.129). The analysis covers Zeek logs and three evidence packet captures recorded on 3 and 5 October 2026.

Zeek logged 165 connections. Of these, 99 are routine background traffic such as time synchronization, DNS, DHCP and browser updates. The remaining 66 belong to four behaviors that were generated deliberately to resemble attack activity. All four originated from the victim host and were directed at the Kali host. Table 1 summarizes them.

Table 1. Summary of findings

ID	Protocol	Observation	Assessment
F-001	HTTP, TCP/8000	30 identical requests to /beacon at 30-second intervals	Beaconing pattern
F-002	DNS, UDP/53	33 queries for random-looking names under malicious-c2.xyz, none answered	DNS tunneling pattern; no data exchange shown
F-003	TCP/5555	One transfer of 52,428,800 bytes (50 MiB) in about 1 second	Large one-way transfer
F-004	TCP/4444	One session on a common attack-tool port carrying the text LAB_TEST three times	Unusual port; harmless content

Because the activity was created intentionally, none of it represents a real intrusion. Each finding is nevertheless assessed as it would be on a production network, and Section 8 lists the corresponding recommendations. All four are rated Medium: the behavior matches known attack techniques, but no harmful content was exchanged and the lab context explains the activity.

2. Scope and Method

The data analyzed consists of Zeek logs (conn, dns, http, dhcp, ntp and packet_filter) covering 3 October 2026 20:38 to 21:00 UTC and 5 October 2026 14:00 to 14:08 UTC, together with three evidence packet captures (beacon, dns and port4444). The Zeek logs were used for connection counts, timing and protocol mix. The captures were used for payloads, protocol headers and exact packet timing. All times in this report are UTC.

Each finding in Section 5 states the observed facts first and the assessment second. The assessment compares the behavior with known attack techniques and notes what further evidence a production investigation would need. Normal protocols are documented in Section 6 so that the four findings can be judged against a baseline.

3. Network Environment

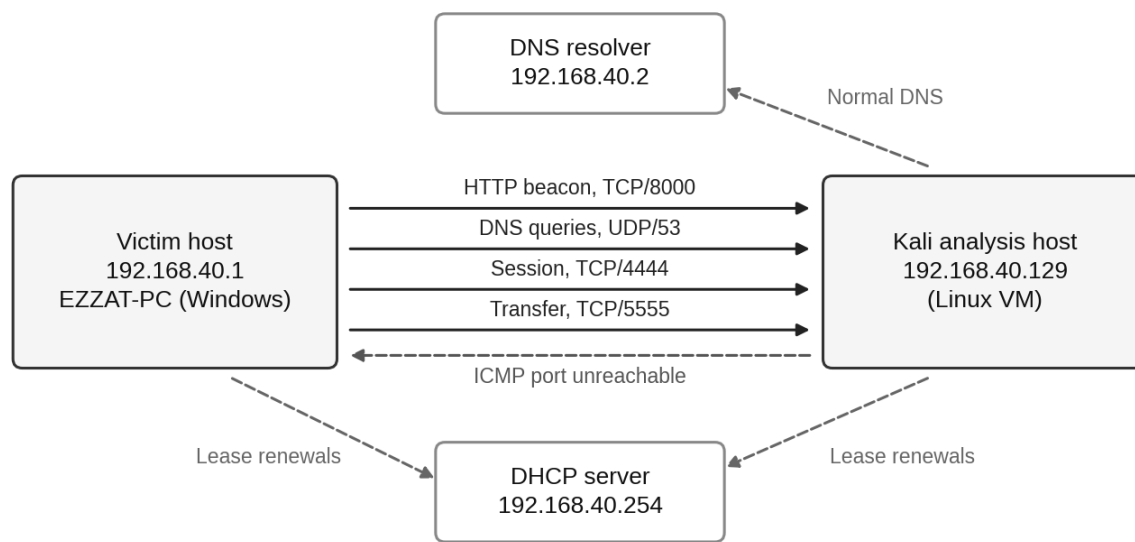


Figure 1. Hosts and traffic observed

Table 2. Hosts

Host	IP address	Hostname	Role
Victim	192 . 168 . 40 . 1	EZZAT - PC	Windows host; originates every unusual flow
Kali analysis host	192 . 168 . 40 . 129	none	Linux VM; runs the listeners on ports 8000, 4444 and 5555
DNS resolver	192 . 168 . 40 . 2	none	Answers normal lookups
DHCP server	192 . 168 . 40 . 254	none	Assigns and renews addresses

Knowing which host starts each connection is important, because attack traffic normally flows from the compromised host to the system that controls it. The host roles follow the lab description. They are consistent with the packets: traffic from the victim carries an IP TTL of 128 and Windows-style TCP options, while traffic from the Kali host carries a TTL of 64 and Linux-style options.

4. Traffic Overview

In this report a connection means one flow record in Zeek's conn.log. For UDP, Zeek groups packets with the same addresses and ports into a single flow, so the counts measure flows, not packets or bytes.

Table 3. Connections by protocol

Protocol / port	Connections	Share	Description	Assessment
UDP/123	58	35.2%	NTP time synchronization	Normal
UDP/53	42	25.5%	DNS	33 unusual, 9 normal
TCP/8000	30	18.2%	HTTP to the Kali host	Unusual
TCP/443	15	9.1%	HTTPS to Mozilla addresses	Normal
UDP/67	5	3.0%	DHCP	Normal
ICMP / ICMPv6	5	3.0%	Ping, port unreachable, multicast reports	Normal
UDP/5355	4	2.4%	LLMNR name queries	Normal
TCP/4444	2	1.2%	Non-standard port	Unusual
IGMP	2	1.2%	Multicast membership	Normal
TCP/5555	1	0.6%	Large transfer	Unusual
UDP/5353	1	0.6%	mDNS	Normal

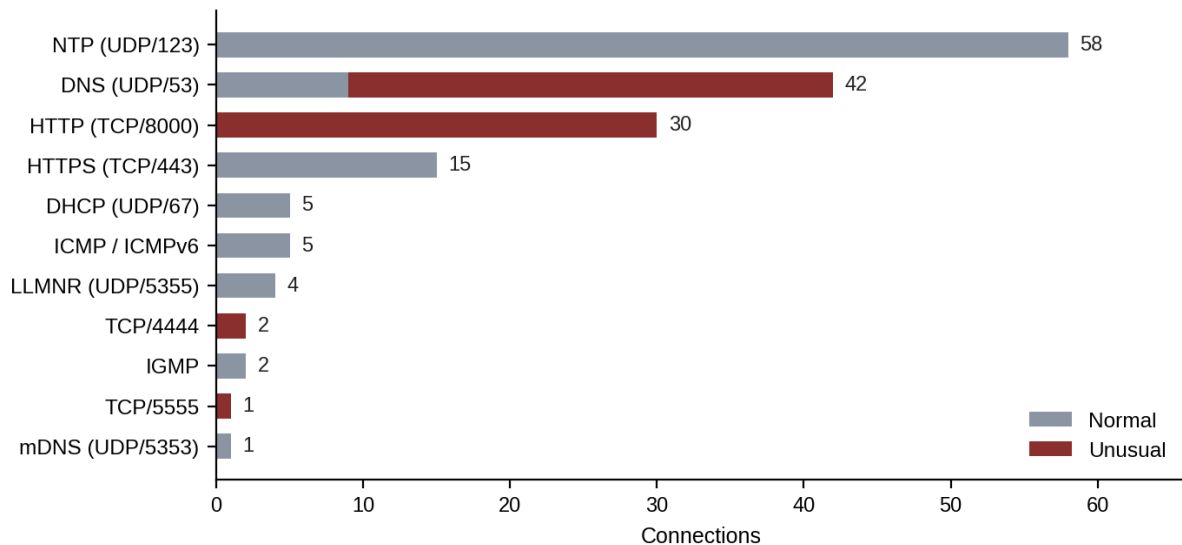


Figure 2. Connections by protocol, unusual traffic shown in dark red

The four behaviors account for 66 of the 165 connections: 30 beacon requests, 33 DNS flows to the Kali host, and 3 connections on ports 4444 and 5555. The single flow on port 5555 carries more than 99.9% of all bytes sent by clients.

5. Findings

5.1 F-001: HTTP Beaconing (TCP/8000)

Between 20:45:49 and 21:00:20 on 3 October, the victim sent 30 requests to `http://192.168.40.129:8000/beacon`. Every request was identical (GET `/beacon` HTTP/1.1, 162 bytes), used the user agent `python-requests/2.34.2`, and carried no referrer or cookies. Each request opened a new TCP connection from a different source port. The server answered every request with the same 335-byte "404 File not found" page, in the format produced by Python's built-in `http.server`, and then closed the connection.

Table 4. Interval between requests (29 intervals)

Minimum	Maximum	Mean	Median	Standard deviation
29.77 s	30.42 s	30.03 s	30.03 s	0.11 s

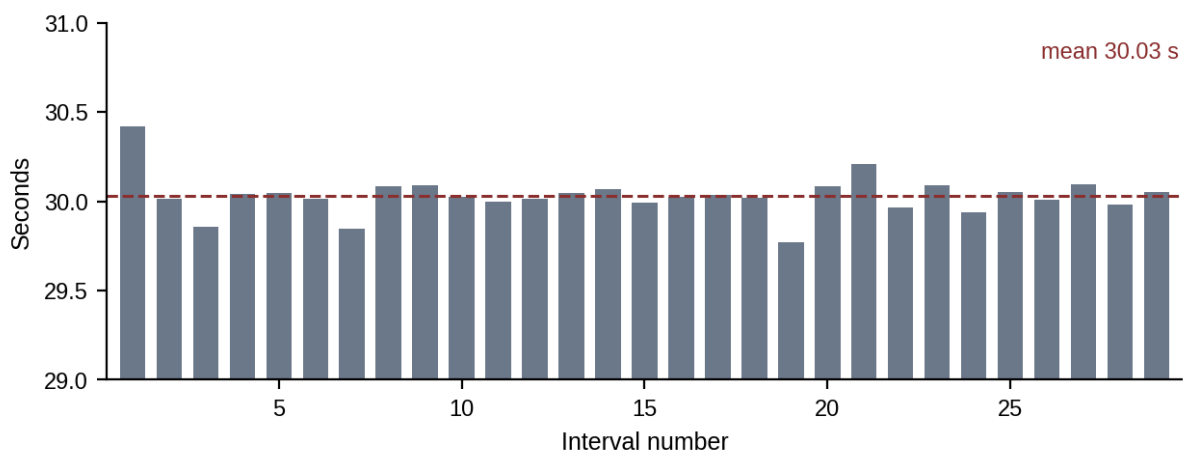


Figure 3. Interval between consecutive beacon requests

Malware commonly contacts its controller at set intervals to ask for instructions. Human browsing is irregular, so a fixed interval is what separates this traffic from normal web use.

Assessment. The mean interval was 30.03 seconds with a standard deviation of 0.11 seconds, a variation of under 0.4%. A fixed cadence, a scripted client and an unchanging destination are typical of beaconing. The server returned only 404 responses, so no tasking or data exchange took place. On a production network, the sending process, the reputation of the destination and a historical baseline would be needed before the activity could be judged malicious.

5.2 F-002: Suspicious DNS Queries (UDP/53)

Between 14:03:41 and 14:06:03 on 5 October, the victim sent 33 DNS queries to `192.168.40.129` on UDP/53 instead of the lab resolver (`192.168.40.2`). The queries consist of 26 address lookups (A and AAAA records) for seven names under `malicious-c2.xyz`, and seven reverse lookups (PTR) of the Kali host's own address. They occurred in seven runs about 10.5 seconds apart. Within each run, the queries repeated every 2 seconds with recursion requested, which is the behavior of the `nslookup` tool. No query was answered. The Kali host replied to each one with an ICMP "port unreachable" message, showing that nothing was listening on UDP/53.

DNS tunneling hides data in the name portion of a query. The attacker's own name server reads the name and can reply with data of its own. For this to work, the query must reach a DNS server that answers. Here the target was the Kali host, which ran no DNS service, so every query failed. The reverse lookups are explained by `nslookup`, which begins each run by looking up the address of the server it was told to use.

Table 5. Names queried under `malicious-c2.xyz`

Name	Length (characters)
random-subdomain	16
l xpht4i9n7vqug5w3mfr	20
t7d2m1i6ycja8hbeg54z	20
f83s5xveji7tn1q2ugbr	20
l7fthbajv4q9p32mcoys	20
m9yjsbv2rk0a734qtngz	20
capf0hr7x65bqn981duw	20

Six of the seven names are 20-character random strings. Each contains 20 distinct characters (4.32 bits of entropy per character), which suggests the names were generated by a script and do not carry encoded data.

Two other DNS observations were reviewed. On 3 October between 20:43:30 and 20:43:35, the Kali host asked the lab resolver for three 10-character hexadecimal names under `example.com` (`8f72a91c32`, `92ad81c72f` and `a7812bc981`) and received empty `NOERROR` replies; Zeek recorded these replies without a query name, whereas the capture shows the names. Lookups for `ads.mozilla.org` and `push.services.mozilla.com` were answered normally and form the baseline.

Assessment. Long, high-entropy labels sent repeatedly to a single domain can indicate DNS tunneling. The labels alone do not prove it, and because no query was answered, no data exchange is shown.

5.3 F-003: Large Transfer (TCP/5555)

Zeek recorded one connection from `192.168.40.1:65137` to `192.168.40.129:5555` at about 20:52:59 on 3 October. It lasted 0.99 seconds and carried 52,428,800 bytes (exactly 50 MiB) toward the Kali host, with no data returned. Zeek logged only 56 packets for this flow, so the byte total is Zeek's count, which it derives from TCP sequence numbers.

Attackers often gather data and send it out in a single burst, so large one-way transfers to unusual ports are examined. In this case the destination is another host inside the lab, not an external address.

Assessment. A large, one-directional transfer to an uncommon port can indicate unauthorized data movement. The exact 50 MiB size is consistent with a test file. On a production network, the transferred content, the user or process responsible and the business justification would need to be established.

5.4 F-004: Connection on Port 4444 (TCP/4444)

At 20:38:41 on 3 October, the victim completed a TCP handshake with 192.168.40.129:4444 from source port 48924. The session lasted 204.9 seconds and was left open. During that time the victim sent the text LAB_TEST three times (9 bytes each, 27 in total), and the server sent no data.

Table 6. Packets in the port 4444 session

Offset	Direction	TCP flags	Content
0.0 s	Victim to Kali host	SYN	none
0.0 s	Kali host to victim	SYN, ACK	none
0.0 s	Victim to Kali host	ACK	none
43.3 s	Victim to Kali host	PSH, ACK	LAB_TEST
76.8 s	Victim to Kali host	PSH, ACK	LAB_TEST
204.9 s	Victim to Kali host	PSH, ACK	LAB_TEST

Port 4444 is the default listener port of the Metasploit framework, and attackers often leave such defaults unchanged. A single session with a few irregular messages is more consistent with a manual test than with automated control traffic.

Assessment. A connection to this port warrants review. Here the payload was plain text with no commands or binary content, and the messages were sent at irregular intervals, which is inconsistent with automated control traffic. The port number alone is not sufficient to establish malicious intent. Zeek recorded this session as two entries, whereas the capture shows one continuous connection.

6. Other Protocols

Table 7. Remaining protocols

Protocol	Observation	Assessment
NTP, UDP/123	57 exchanges with three time servers (196.44.136.165, 196.10.55.57, 196.10.54.57) at about 30-second intervals; NTP version 4, stratum 2	Normal
DHCP, UDP/67	Five lease renewals by 192.168.40.129 and 192.168.40.1 (EZZAT-PC), about 900 seconds apart on a 1800-second lease	Normal
HTTPS, TCP/443	15 flows from the Kali host to 199.232.81.91 and 34.107.243.93, the addresses returned for the Mozilla lookups; content encrypted	Normal
LLMNR and mDNS	Queries for ezzat-pc at 20:42:17 and 20:57:26 on 3 October, and one mDNS query at 20:49:39	Normal Windows name resolution
ICMP	One ping at 14:02:57 on 5 October, and 33 port-unreachable replies from the Kali host to the DNS queries in F-002	Normal; supports F-002
IGMP and MLD	Multicast membership reports from the victim at 20:42:17 and 20:57:26 on 3 October	Normal

The NTP, DHCP and HTTPS traffic follows the regular timing and destinations expected from a Windows and a Linux host. The ICMP port-unreachable replies are significant only because they confirm that the DNS queries in F-002 went unanswered. LLMNR is normal on Windows, but it can be abused on real networks to capture credentials, and many organizations disable it.

7. Timeline

Table 8. Sequence of events

Time (UTC)	Protocol	Event
2026-10-03 20:38:41	TCP/4444	Session to the Kali host begins
2026-10-03 20:43:30	DNS	Hexadecimal-name lookups under <code>example.com</code>
2026-10-03 20:45:49	HTTP	First <code>/beacon</code> request
2026-10-03 20:52:59	TCP/5555	50 MiB transfer
2026-10-03 21:00:20	HTTP	Thirtieth and last beacon request
2026-10-05 14:02:57	ICMP	Ping from the victim to the Kali host
2026-10-05 14:03:41	DNS	First <code>malicious-c2.xyz</code> run begins
2026-10-05 14:06:03	DNS	Last captured query

8. Conclusions and Recommendations

The captured traffic contains four behaviors that match common attack techniques: periodic HTTP beaconing, DNS queries resembling tunneling, a large one-way transfer and a session on port 4444. All four were generated deliberately, none exchanged harmful content, and the remaining traffic behaved normally. If the same patterns appeared on a production network, the actions in Table 9 would apply. The priorities reflect how directly each control would stop or reveal an active communication channel: blocking the DNS domain and alerting on the beacon address address the channels themselves, while the remaining items improve detection and reduce exposure.

Table 9. Recommended actions

Priority	Protocol	Recommendation
High	DNS	Block or sinkhole <code>malicious-c2.xyz</code> ; permit DNS only to approved resolvers and alert on queries sent to other internal hosts
High	HTTP	Alert on fixed-interval requests to the same address, especially from script clients such as <code>python-requests</code>
High	TCP/5555	Review large transfers to uncommon ports and confirm what was sent
Medium	TCP/4444	Alert on connections to port 4444 and identify the program at both ends
Medium	DNS	Detect long, random-looking names and repeated failed lookups
Medium	LLMNR	Disable LLMNR and NetBIOS name lookup where they are not needed
Low	Zeek	Confirm the sensor sees both directions of traffic so that counts and byte totals are complete

9. Limitations

- Zeek often saw only one direction of a flow, so some counts, connection states and byte totals are partial.
- Host roles come from the lab description, supported by TTL values and TCP options; no host forensics were performed.
- HTTPS content is encrypted, so only addresses and counts were reviewed.

Appendix A. Wireshark Display Filters

Purpose	Filter
Beacon requests	<code>http.request.uri contains "beacon"</code>
All HTTP to the Kali host	<code>tcp.port == 8000</code>
Suspicious DNS	<code>dns.qry.name contains "malicious-c2"</code>
Reverse lookups sent to the Kali host	<code>dns.qry.type == 12 && ip.dst == 192.168.40.129</code>
ICMP port unreachable	<code>icmp.type == 3 && icmp.code == 3</code>
Port 4444 session	<code>tcp.port == 4444</code>
Large transfer	<code>tcp.port == 5555</code>
Test payload	<code>frame contains "LAB_TEST"</code>

Appendix B. Terms

Term	Meaning
Beaconing	A program contacting a controller at regular intervals
DNS tunneling	Carrying data inside DNS queries and replies
TTL	A packet counter whose default differs by operating system (128 for Windows, 64 for Linux)
Zeek	A network monitor that writes logs of connections and protocols